Dams mark multi-year drought vulnerability rather than buffer it in Chile's megadrought

Francisco Zambrano Bigiarini[a,*]

[a]*Facultad de Medicina Veterinaria y Agronomía, Chile, Universidad de Las Américas, Santiago, Chile,*

Abstract

Expanding reservoir storage is the reflexive response to intensifying drought, yet whether dams build resilience or merely mark existing vulnerability has remained untested for want of a credible counterfactual. We treat reservoir presence as a basin-level intervention in Chile's megadrought, comparing how matched dammed and undammed basins respond to the same deficit at annual to multi-year accumulations rather than seasonal operation. Across streamflow drought transmission, cropland change, and consumptive water rights, the reservoir effect brackets zero once siting is matched out. The cleanest test compares reaches below each dam with unregulated reaches above: the apparent buffering is as strong upstream as downstream, the signature of aridity, not regulation. The storage band drifts downward, mirroring the independent decline in unregulated streamflow. Reservoirs mark vulnerability established before our record; through the megadrought we detect neither added demand nor buffering, within bounds that admit volumetrically material effects. Adaptation must shift toward managing demand.

Keywords: reservoir effect, drought propagation, socio-hydrology, causal inference, megadrought, Chile

As warming intensifies drought across many regions[1,2], governments instinctively build more dams. Reservoirs are the dominant structural response to drought, and in the short term they work: more storage raises availability and lowers the frequency, severity and duration of shortage[3]. Yet a socio-hydrological view warns the same infrastructure can be self-defeating over a dam's planning horizon. Greater supply invites expansion, so demand grows to meet and then exceed each increment of storage (the supply-demand cycle); and prolonged reliance on stored abundance erodes the incentive to adapt, magnifying the damage when a drought finally outlasts the reservoir (the reservoir effect)[3]. Consistent with this, global demand has

*Corresponding author
Email address: frzambra@gmail.com (Francisco Zambrano Bigiarini)

outpaced storage for decades, returns of new dams are diminishing[4], and most reservoirs now fail to refill[5]. Whether these feedbacks dominate, or are outweighed by the protection dams provide, remains debated[6].

The obstacle is that the reservoir effect for drought has been described far more than measured. Reservoirs are built where irrigation demand and aridity already concentrate, so dammed basins differ systematically from undammed ones before any drought strikes. The short-term hydrological benefit is documented[7,8], though regulation can also redistribute scarcity in space and time[9]. But the long-term demand side is far less constrained, and separating a reservoir effect from siting requires a defensible counterfactual, what a drought's impact *would have been* had the reservoir not been built[3]; prior dammed-versus-undammed and before-after comparisons cannot supply it.

Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an uninterrupted run of dry years shading into structural aridification, a drying with a documented anthropogenic component[10–14]. Because the 1981 Water Code makes water-use rights perpetual private property that the state cannot readily curtail, the policy reflex has been supply-side: new dams and inter-basin transfers, alongside market incentives that concentrated water in high-value export crops[15–17]. In the snow-fed basins of central Chile this demand already outruns supply in roughly three years of every ten[18]. Chile thus exhibits every ingredient of the reservoir effect: extreme persistent forcing, a supply-side response, and demand expansion under weak adaptive feedback, with most central Chilean farmers meeting a decade of drought through short-term agronomic cutbacks rather than durable adaptation[19].

Here we ask whether reservoirs build resilience or merely mark vulnerability under prolonged drought. We treat the presence of a major reservoir as a basin-level intervention and construct the missing counterfactual in two steps. First, we compare each dammed basin only against undammed basins sharing its climate and landscape, conditioning on the drought actually experienced. Second, within each dammed basin we compare regulated reaches below the dam against physically unregulated reaches above it, which share climate, geology and siting but not the reservoir. Throughout, outcomes are evaluated at annual to multi-year accumulations, the timescale of structural drought, not of seasonal operation. Together these turn the reservoir-effect hypothesis into a falsifiable, causally framed test in the setting where it matters most. The answer travels: drying Mediterranean-climate regions from western North America to Iberia and Australia face the same choice between more storage and demand reform[20,21].

Results

Reservoirs sit where water is already scarce

Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster in the arid centre and north, so a raw comparison would confuse the reservoir with its location. We separate them with the design in Figure 1. First, we compare each dammed basin only with undammed basins sharing its climate, size and elevation, keeping 21 of 24 dammed basins against an effective 91 of 244 controls (Supplementary Figs. S4 and S5), and we condition on the drought each basin experienced, comparing impact per unit of deficit. Second, within each dammed basin we compare regulated reaches below the dam against unregulated reaches above it. A genuine reservoir effect must appear in these comparisons; in none does it.

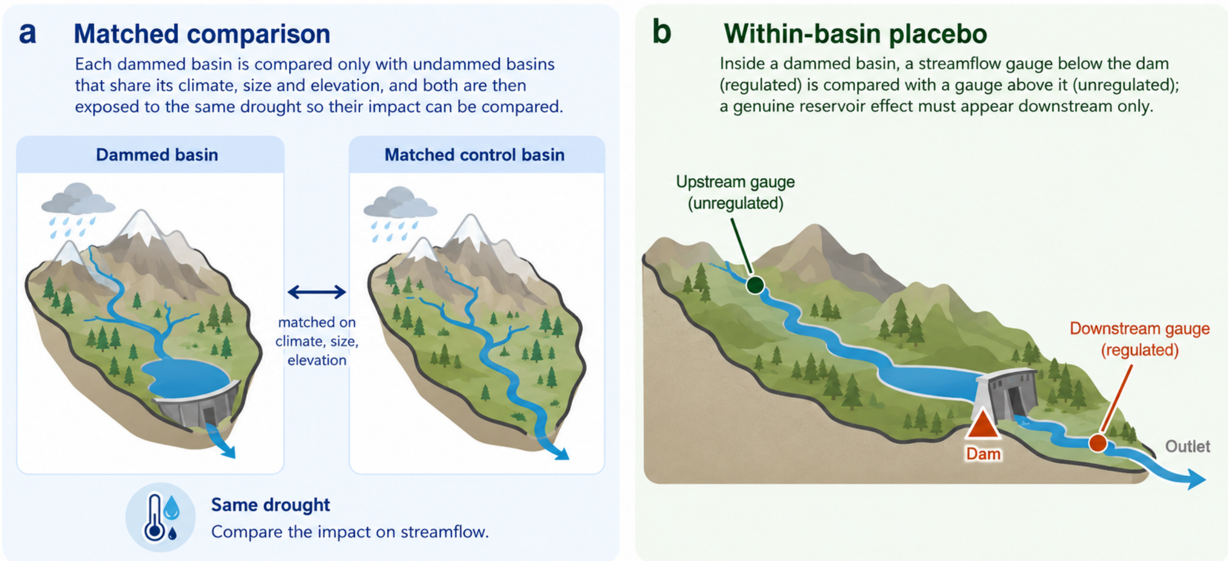

Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin is compared only with undammed basins that share its climate, size and elevation, and both are then exposed to the same drought so their impact can be compared. (b) Within-basin placebo: inside a dammed basin, a streamflow gauge below the dam (regulated) is compared with a gauge above it (unregulated); a genuine reservoir effect must appear downstream only. Schematic; the full study-area map and matching diagnostics are in Supplementary Figs. S4 and S5.

The apparent buffering is the location, not the dam

The most direct test is on streamflow. Meteorological drought (Standardized Precipitation Evapotranspiration Index of 12 months, SPEI-12) propagates into streamflow drought (Standardized Streamflow Index of 12 months, SSI-12) more gently in dammed basins, the visual signature of buffering (Figure 2, panel a). But this flattening tracks where dams sit, not what they do. If the dam were doing the buffering, transmission would flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches of the same basins (differential slope −0.20 upstream versus −0.16 downstream; Figure 2, panel b). The national slope gap of −0.18 accordingly collapses to about −0.05 once dammed and control basins are compared within the same climate region, and in the high-supply south, where buffering should be easiest, the upstream and downstream gaps are identical (−0.036 versus −0.034). None survives randomization inference, which asks how often reshuffling the dammed label among comparable basins produces a gap this large by chance ($p_{\text{perm}} = 0.39$). Nor is the pooled slope hiding early buffering: at drought onset (2010–2014, storage near-full) the upstream and downstream gaps are identical (−0.43 both), and the later weakening is sharpest in the unregulated placebo itself, forcing, not depletion (Supplementary Table S24). The apparent buffering is therefore the between-region aridity gradient, the signature of a siting confound, not regulation; a decomposition confirms the signal dies only when each climate region has its own baseline transmission (Supplementary Fig. S3 and Table S4). Hard-balancing aridity gives the same verdict without parametric adjustment: with log-aridity fully balanced (effective control sample 19), the apparent buffering vanishes (intent-to-treat −0.02, upstream −0.05, permutation $p > 0.85$; Supplementary Table S17), exactly what a siting confound predicts. The parametric adjustment it replaces interpolates within support, every dammed basin inside the control aridity range (Supplementary Fig. S8). Nor is the null a standardization artifact: SSI is normalized per gauge, which would divide out dam-induced variance compression, but on log flow the gaps stay null (downstream +0.04, upstream +0.10, permutation $p \geq 0.42$) and downstream relative variability is uncompressed ($p = 0.43$; Supplementary Table S19). A direct geomorphic metric agrees: local relief, separating bedrock from alluvial reaches, does not predict control transmission, leaves the placebo unchanged, and shows no buffering among alluvial units (Supplementary Table S22). Upstream gauges sit higher, but the 0.54 km up/down elevation gap times the control transmission sensitivity (−0.21 per km) already over-explains the observed −0.04 difference and cannot manufacture a downstream-specific effect, leaving a residual downstream-specific buffering of at most about 0.07 (13% of the 0.53 downstream baseline), well below the 46 to 59% detectable (Supplementary Table S2).

No reservoir effect survives across any operational pathway

The same answer holds across the other pathways (Figure 3, Table 1). Each effect is the differential response of dammed and matched-control basins to the same deficit, each estimate scaled by its own standard error, so a value inside ±2 is indistinguishable from no effect. Every estimate that carries evidential weight brackets zero: cross-sectional cropland-area expansion (−0.69 ha km^{-2}, 95% CI $[-2.4, 1.1]$) and its forcing-conditioned slope gap ($p_{\text{perm}} = 0.91$). The evapotranspiration pathway is relegated to the Supplementary Information: whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the post-hoc orchard stratum relies on MODIS (Moderate Resolution Imaging Spectroradiometer) MOD16 product, whose documented

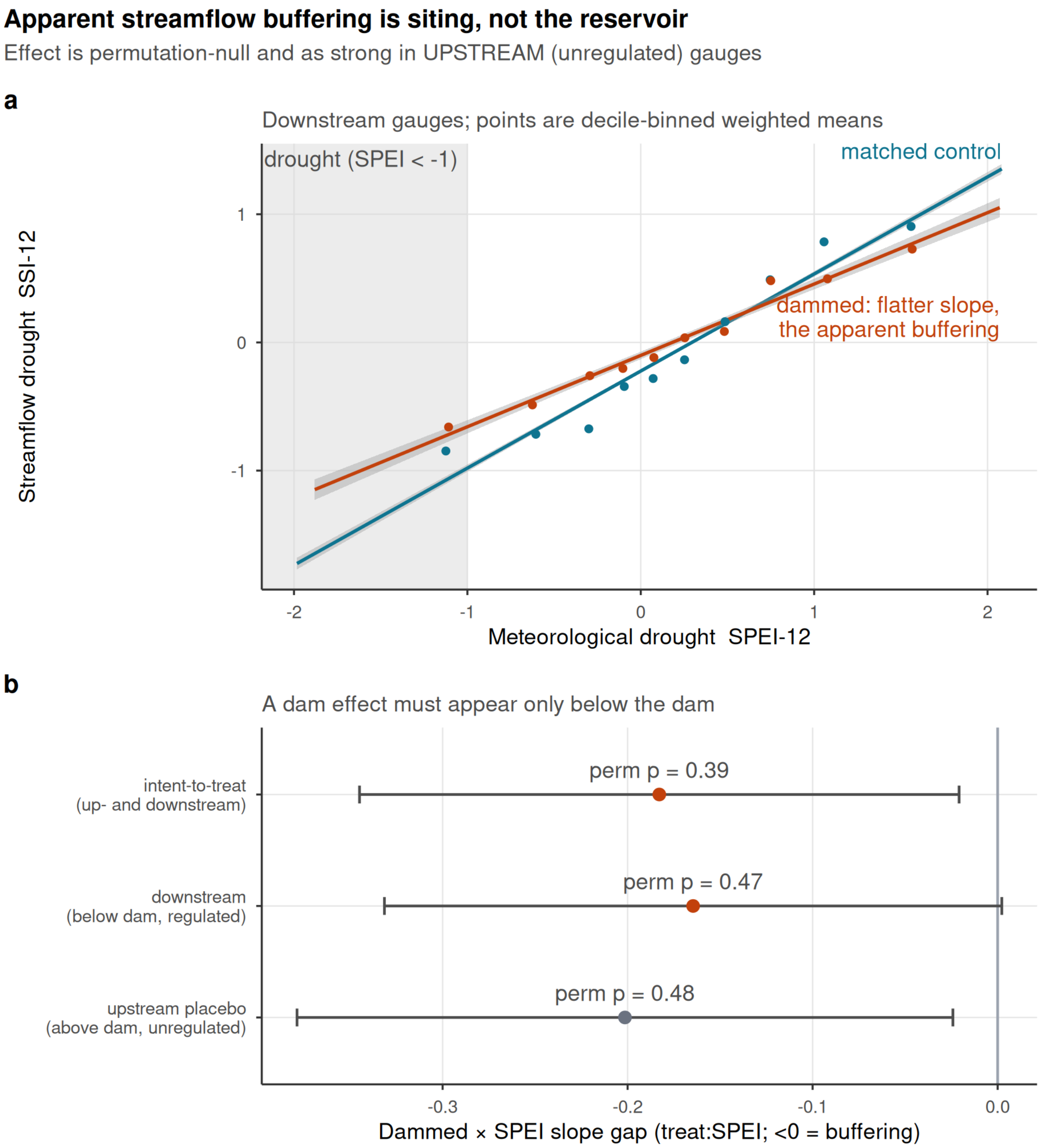

Figure 2: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo (above-dam, unregulated) is as strong as downstream, so the attenuation is siting/aridity, not the dam.

bias toward zero over irrigated land, a product property identified independently of our results, disqualifies its null (Supplementary Table S21, Fig. S7).

The null is informative but its power is concentrated in the streamflow pathway. With about 21 dammed clusters the design cannot claim exact equivalence to zero but rules out buffering above 46 to 59% of baseline, and a positive control confirms it: injected buffering is recovered one-to-one and detected beyond that threshold (Methods; Supplementary Fig. S2 and Tables S1, S3). The cropland design is far weaker, with a minimum detectable effect orders of magnitude above baseline (Supplementary Table S2), so its null corroborates the streamflow result rather than bounding its pathway.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs (including the water-rights induced-demand test) and forcing-interacted DiD slope gaps (treat:SPEI). The DiD row and water-rights accrual are judged by randomization-inference p-values (p), the design's primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. Evapotranspiration outcomes are relegated to the Supplementary Information (Table S21, Fig. S7): whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the orchard rows rely on MOD16, whose documented underestimation over irrigated land biases them toward zero.

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Cropland-area expansion	ha km^-2 added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Water-rights accrual	rights / 100 km^2 added	20.036	[4.02, 36]	0.52	null
Forcing-interacted DiD (slope-gap)	Cropland area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null

Reservoirs mark prior vulnerability; no megadrought divergence is detected

Dammed basins carry the vulnerability reservoirs are blamed for concentrating: about 4.5 times more cropland than matched controls (10.8% versus 2.4% of basin area; Figure 4, panel a). This gap is a *siting* signature, stable across the 2000–2024 record. A dynamic event study finds the differential trend flat before 2010 (cluster-robust $p = 0.65$, permutation $p = 0.72$) with no post-2010 divergence detected ($p = 0.81$; Figure 4, panel b). Both are bounded nulls, not confirmations of parallelism: the pre-trend test detects only divergence accumulating to about half the control cropland level by 2009 (the observed trend is a seventh of that), and the permutation envelope widens after 2015, so the late panel cannot rule out substantial divergence (Supplementary Table S14). The null is metric-invariant: on log cropland share the slope gap is likewise permutation-null ($p = 0.37$; Supplementary Table S20). Reservoirs mark where cropland already was; within these bounds they neither expanded it nor changed its drought response.

The same holds for the most direct demand measure. In the Chilean Water Directorate (DGA) registry of consumptive water rights, dammed basins appear to accrue rights faster than controls, on both the outlier-robust count (+20 per 100 km^2; Figure 3, Table 1) and the winsorized volume (+2.5 l/s per km^2;

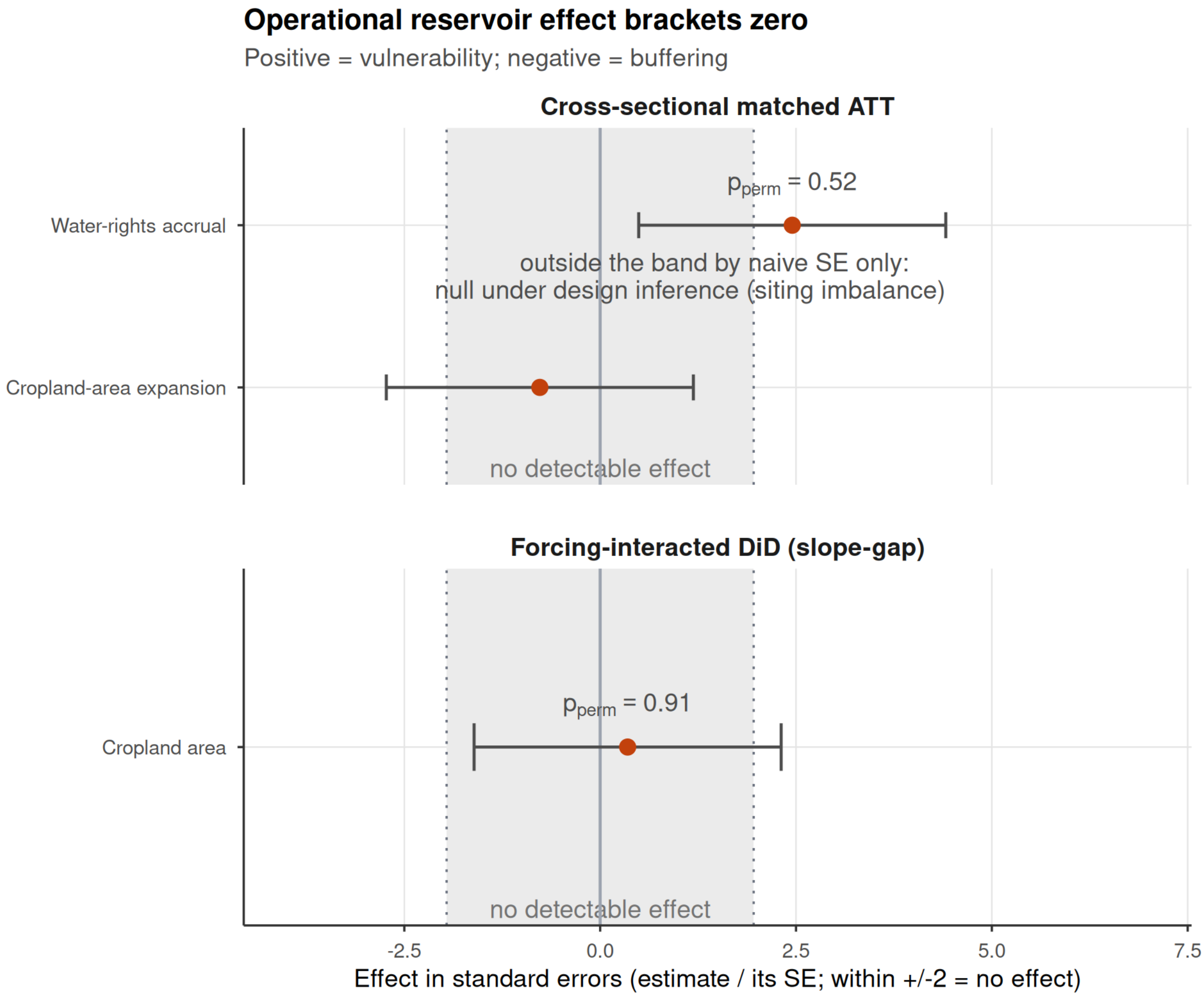

Figure 3: Dammed-versus-control effect across the estimators, each expressed uniformly in its own standard errors (estimate divided by its SE) so the axis metric is identical for every row; an estimate inside +/-2 is indistinguishable from no effect. Rows are grouped by inference regime: cross-sectional matched ATTs (top) and forcing-interacted DiD slope gaps (bottom). Bars span +/-1.96 SE (the 95% reference interval) and dotted lines mark +/-1.96; a positive effect would indicate induced-demand vulnerability, a negative one operational buffering. Rows judged by randomization inference are annotated with their p (the design's valid inference at ~21 clusters, since the +/-1.96 SE guides over-reject there). Under that valid inference every estimate brackets zero. Evapotranspiration outcomes are relegated to the Supplementary Information (Table S21) as product-limited and confounded. Water-rights accrual sits outside +/-1.96 by its naive standard error yet is null by the design (p__perm = 0.52; the apparent induced demand lies inside the permutation null expected from the dammed basins' unbalanced aridity, Supplementary Fig. S6).

Supplementary Fig. S6), each with an analytic interval excluding zero. This is the naive-analysis trap the design catches: dammed basins are more arid, arid basins carry denser water-rights activity, and the imbalance drives the gap. The count does not survive randomization inference ($p_{\text{perm}} = 0.52$; Supplementary Fig. S6) and remains null under a fully non-parametric match on the aridity-overlap subset ($p_{\text{perm}} = 0.42$; Supplementary Table S7). The winsorized volume is more delicate: stable across thresholds and physically validated (Supplementary Table S8), its overlap-subset permutation p falls to 0.005, but the residuals are spatially autocorrelated (Supplementary Table S9) and under the governing cuenca-block permutation it is not significant ($p_{\text{perm}} = 0.077$, cuenca-clustered interval $[-0.3, 4.1]$). Ranked raw, uncapped volumes give the same verdict: the design rank contrast is null under cuenca-block permutation ($p = 0.15$; Supplementary Table S18). Even on water rights, no induced demand is robustly attributable to the reservoir once siting and spatial structure are accounted for.

Storage falls; a change in refill amplitude cannot be resolved

If reservoirs neither buffer nor amplify drought, what matters is how much water there is to store (Figure 5). Over 2005–2024 the storage band drifts downward in step: peak storage falls by 1.3 percentage points of capacity per year ($p = 0.013$) and the trough almost identically (1.2 pp yr^{-1}, $p < 10^{-4}$), while the seasonal amplitude shows no detectable trend (-0.06 pp yr^{-1}, $p = 0.91$; Supplementary Table S5). That amplitude interval is wide: whether refill amplitude, the buffering capacity, has degraded cannot be determined, and we draw no conclusion about it. The decline is a level shift at the megadrought onset rather than a secular line (peak 87% of capacity before 2010, 67% after; trough 40% to 25%; Supplementary Table S5). That decline is commensurate with the supply reduction measured independently in the matched controls: unregulated control streamflow fell about 2.9% per year over 2005–2020 (dammed 3.4%), consistent with documented megadrought runoff deficits[10,22] and comparable to the ~1.4% per year decline in peak storage. Storage is a mass balance, so the band alone cannot separate falling inflows from sustained releases; the supply reading rests on that independent streamflow decline. Our direct demand outcomes point the same way: neither cropland area nor water-rights allocation shows expansion (Figure 4, Table 1; Supplementary Fig. S6)[4,5].

Discussion

Across a matched national design and multiple estimators, the operational effect of reservoirs on drought brackets zero once siting is matched out. Dammed basins hold about 4.5 times more cropland than matched controls, but that gap is fixed across the record, with parallel pre-trends and no detected megadrought

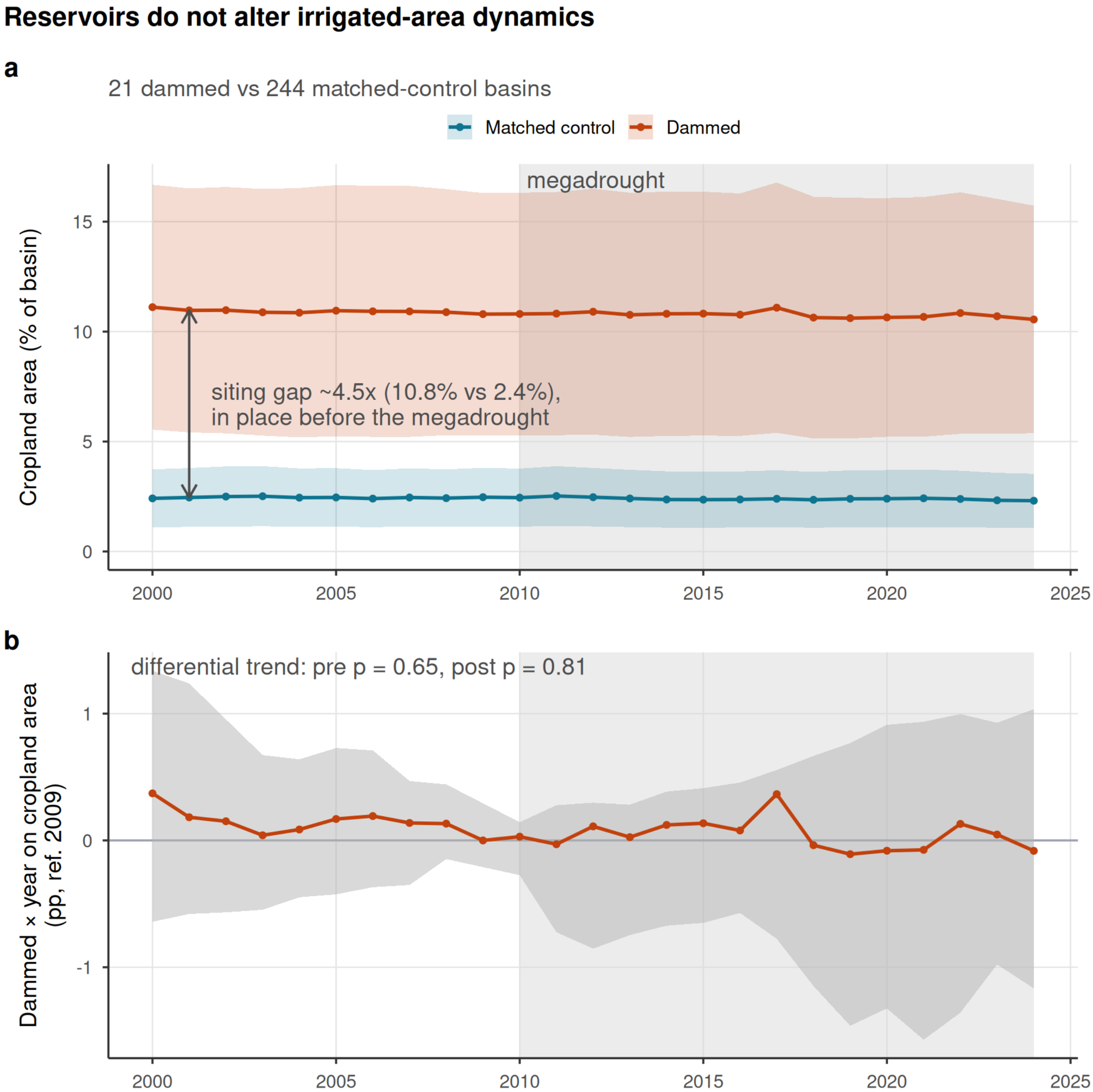

Figure 4: Observed cropland-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, MapBiomas class 18) with 95% bands (weighted mean +/- 1.96 SE across basins); the y-axis is anchored at zero. Dammed basins hold ~4.5x more cropland (10.8% vs 2.4% of basin area) but the trajectories evolve in parallel. (b) Dynamic event study (dammed x year, reference 2009): the grey band is the 95% randomization-inference null envelope (2.5th to 97.5th percentile of the dammed x year coefficients under treatment permutation within Köppen x aridity strata); coefficients inside the band are indistinguishable from no divergence, and cluster-robust intervals are omitted as anti-conservative at ~21 treated clusters. The differential pre-trend is flat (pre-2010 trend-slope cluster-robust p = 0.65, permutation p = 0.72) and no post-2010 divergence is detected (post p = 0.81); a linear trend-slope test is used because the joint event-study Wald over-rejects at ~21 treated clusters. The null envelope widens after 2015, reflecting declining power in the later panel, so the flat post-2015 trajectory is a bounded null: divergence smaller than the envelope cannot be ruled out. The 2010-2024 megadrought is shaded in both panels.

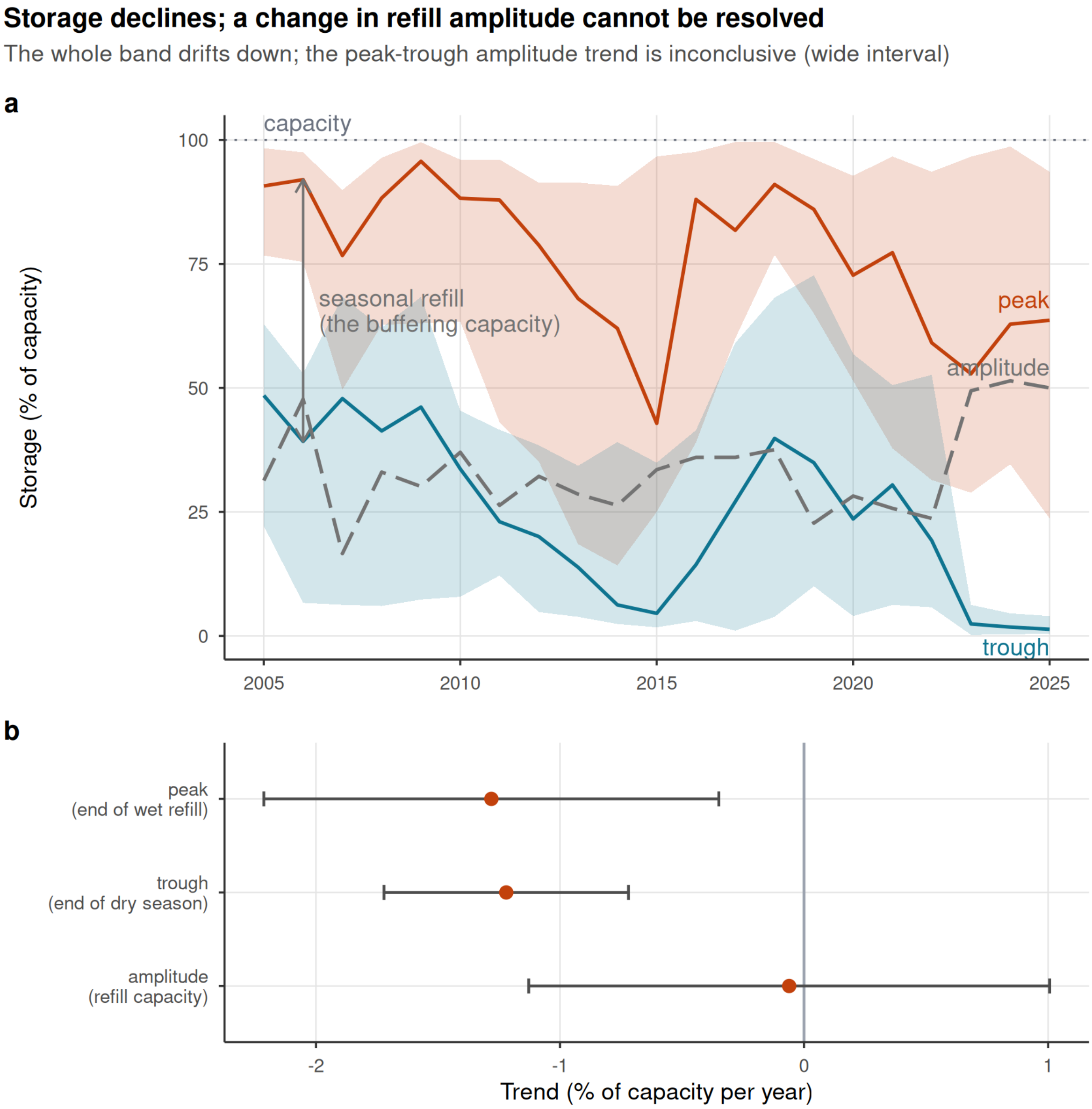

Figure 5: Reservoir storage declines; a change in seasonal refill amplitude cannot be resolved. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid lines) with interquartile ribbons across reservoirs, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the whole band drifts down over 2005-2024 while the amplitude does not narrow. Central lines are the realized medians, not linear fits; the net trend is quantified in (b). (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE): peak and trough both decline while the amplitude trend is not distinguishable from zero (a wide interval, so statistically unchanged rather than proven constant), consistent with a supply-side level decline rather than refill/buffer degradation. Descriptive: pooled within-reservoir trend, no control group.

divergence. One reading fits: reservoirs mark where water-dependent vulnerability already concentrated rather than make it. This is a finding, not a failure of power: the apparent effect has an identified cause and the positive control shows the design detects buffering. The placebo is the cleanest falsification: a regulation effect must appear only below the dam; it does not, and what remains is the aridity gradient.

The corollary is transferable: dammed-versus-undammed or before-after contrasts inherit the full siting confound[7,9,23,24]. The upstream placebo ports to regulated rivers with gauges above and below a dam, most directly to short, steep catchments like Chile's; in large low-gradient basins, floodplain storage and transmission losses would blur the contrast; porting it would need routing-corrected lags or naturalized-flow counterfactuals.

If reservoirs neither buffer nor amplify multi-year drought operationally, what matters is the water available to store. The storage band steps down at the megadrought onset, peak and trough together, while the amplitude trend is too imprecise to determine whether buffering capacity degraded. The powered result is the level decline, the expected mass-balance outcome of persistent inflow deficits; descriptive and confounded with management change, it supports no causal attribution but matches the independent decline in unregulated control streamflow. Read against Chile's structural aridification[10,12,22], accelerating Andean warming and snow loss now cutting into the snow-fed streamflow these basins depend on[25], and the global pattern of reservoirs failing to refill[4,5], the binding constraint is shifting toward supply.

These findings give the contested socio-hydrological *reservoir effect* and *supply-demand cycle*[3,6] a credibly identified test, and find the protective function over-attributed to storage. Because treatment is time-invariant, we test the spatial imprint of these feedbacks, not the longitudinal loop. The 4.5-fold cropland gap could be the equilibrated legacy of demand induced at construction, so our null covers the megadrought era; historical induced demand is testable only with staggered commissioning. And the record opens in the scarcity phase, after any expansion under perceived abundance, so it bounds drought-era expansion, not the cycle as a whole. Within scope the nulls refine the theory: the cycle leaves no operational imprint during drought, so its mechanisms likely act through siting and construction rather than operation, a distinction current models omit. How far the demand-side null travels depends on governance: the siting lesson holds wherever dams are placed non-randomly; prior appropriation, which forfeits unused rights, may amplify induced demand while adaptive allocation curtails it. Whether the null is general or the mark of a fully allocated market we cannot adjudicate: rights kept accruing in dammed basins, but systems with water still unallocated could express an induced demand that ours cannot.

Chile's 1981 Water Code makes water-use rights perpetual private property, so the reflexive response has been supply-side, new dams and inter-basin transfers[11,15]; the 2022 reform moves toward the adaptive allocation our results argue for[26] but postdates our window. Under structural aridification, building more storage is a supply-blind response to a supply problem: the buffering dams provide is a property of where they sit, not of what they do[21]. Adaptation must shift toward demand management, reallocation and managed aquifer recharge[27–29].

Several limitations bound these conclusions, each quantified in the Supplementary Information. The result is a bounded null: the design excludes streamflow buffering above 46 to 59% of baseline, demand-side outcomes are far less powered and the pre-trend test power-limited (Supplementary Tables S2, S14), so causal weight rests on the placebo, demand nulls corroborating; all survive the hard-balanced match (Supplementary Table S17). The registry measures paper rights, not wet water: Chilean basins are widely over-allocated and granted rights can sit unused or hoarded[15,16], so under drought, when physical availability binds, the accrual we test is an entitlement response that non-use could dampen; cropland is the physical check. Both metrics also miss intensification within an unchanged footprint; central Chile's documented shift toward thirsty export fruit[16] is exactly such a transition, invisible to both; a consumption-grade test awaits thermal ET or crop-transition maps. Groundwater substitution is bounded, dammed and control water tables falling at indistinguishable rates across 213 DGA wells despite severe regional drawdown[30]; monitored wells need not sample the orchard footprints where pumping concentrates, so unmonitored drawdown could sustain a historically reservoir-induced footprint (Supplementary Table S10). In that reading the reservoir effect is not absent but displaced to the aquifer, a construction-era legacy persisting on a pool the design cannot see; the well bound and flat cropland trajectories argue against substitution at scale but do not close the pathway. The snowmelt confound, acute in these snow-fed Andean basins[31,32], is tested on the differential estimand and excluded across four specifications (Supplementary Table S12). The geomorphic check leaves the placebo unchanged, though 2.5 km relief cannot resolve aquifer storage itself, a residual caveat (Supplementary Table S22). The placebo also spans land-cover and abstraction contrasts, downstream valleys more cultivated, upstream reaches carrying unrecorded diversions; gauge fixed effects absorb these level differences, and drought-intensified upstream abstraction would steepen upstream transmission, working against the observed upstream attenuation, not producing it. Administrative closure cannot force the demand null: treated basins entered the drought with higher allocated stock yet accrued rights 2.7 times faster, and parallel control cropland contradicts collapse prevention (Supplementary Table S15). No matched pair shares a watershed; outbound transfers would shrink the gap and imports would mimic downstream-specific buffering, which is absent (Supplementary Table S15); irrigation-only restriction leaves the result unchanged (Supplementary Table S11). MOD16's bias toward the null excludes the ET pathway from evidence (Supplementary Table S21); the demand conclusion rests on cropland and the registry, with thermal ET as future work. Finally, the 12-month accumulation targets multi-year vulnerability, not the sub-seasonal buffering reservoirs also provide, which we neither test nor dispute. The fleet is partly seasonal by design, the median treated reservoir storing 0.3 years of downstream flow, yet seven of 17 hold at least half a year, four exceed a year, and none empties seasonally (Supplementary Table S23): carryover was physically available, so its absence in outcomes is not a design tautology. Two extensions can test what our window cannot: a staggered event study of cropland around historical commissioning would probe construction-phase induced demand; a rule-curve quasi-experiment on release records, once public, would isolate operation.

At the multi-year timescale of structural aridification, then, reservoirs neither create resilience nor postpone vulnerability operationally: storage neither buffers nor amplifies the drought signal beyond what siting implies, while the water available to store declines. The nulls are bounds, not zeros; effects inside them can be volumetrically material (Supplementary Table S16). Resilience to a falling supply ceiling will come from confronting demand and allocation, not more impoundments.

Methods

Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates, over 2005–2024, the window in which reservoir storage, meteorological forcing, and satellite outcomes jointly exist. The design is a matched comparison of dammed versus undammed sub-watersheds, with reservoir presence treated as a basin-level intervention. Because the reservoirs in most treated basins (18 of the 24) were commissioned before the storage record begins, treatment is effectively time-invariant over the window and a staggered-adoption (commissioning) event study is infeasible; identification therefore rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited*, that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir *operation* from reservoir *siting*[3].

Spatial units and treatment assignment

The analysis grain is the DGA sub-watershed (467 units), chosen over the coarser watershed grain (101 units) for power and common support, reservoirs sit in Chile's largest main-stem basins, and the finer grain supplies far more controls inside the treated size range (detailed in the archived analysis repository). A sub-watershed is treated if it contains 1 of the 26 monitored reservoirs by point-in-polygon assignment, giving 24 treated units. The reservoirs are predominantly irrigation (20 of 26; three hydropower, two potable supply, one mixed), so the treated set is dominated by the use type built to buffer agricultural drought, and the headline streamflow result is unchanged when restricted to irrigation-only treated basins (Supplementary Table S11). An untreated sub-watershed that lies within a watershed containing a reservoir is flagged as contaminated (it shares the regulated main stem and is not a clean counterfactual) and removed (45 units), leaving 398 clean controls. This contamination rule defuses the most direct downstream-spillover (SUTVA: Stable Unit Treatment Value Assumption) violation; residual adjacency and shared-aquifer spillovers between neighbouring sub-watersheds are carried as an identification caveat and addressed with watershed-level checks: where spatial dependence is present, inference is re-run with watershed-block permutation and watershed-clustered intervals (Supplementary Table S9), and no matched treated-control pair shares a watershed (Supplementary Table S15).

Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (the DGA record runs January 2005 to April 2026; we analyse the complete calendar years 2005–2024) with a per-reservoir capacity column (maximum level in hm^3); storage and capacity share units (hm^3), so percent-of-capacity (`100 × level / maximum level`) is the cross-reservoir-comparable state variable. Static attributes (commissioning year, use, size class) come from the national dam registry.

Meteorological forcing. The Standardized Precipitation-Evapotranspiration Index (SPEI) at a 12-month accumulation, derived from Climate Hazards Center InfraRed Precipitation version 3 (CHIRPS) and Climate Hazards Center InfraRed Temperature with Stations (CHIRTS) and standardized on the 1991–2020 climatological baseline[33], is the primary exogenous dose; CHIRPS-derived standardized indices are validated for Chilean drought monitoring, including ENSO-modulated variability across the country[34–36]. We use the 12-month accumulation because it matches the annual, snowmelt-fed hydrological cycle of central Chile (winter precipitation stored as snow and released through the growing season) and so captures the inter-seasonal carryover that irrigation reservoirs are built to bridge, at the scale at which multi-year vulnerability (rather than sub-seasonal operation) is expressed. The treatment estimates are insensitive to this choice: re-estimating the forcing-conditioned effect with 6- and 24-month accumulations leaves it materially unchanged (archived analysis code). By construction the 12-month window does not resolve short-term (1–3 month) operational buffering, which reservoirs are documented to provide[7,8] and which we flag as a boundary of our conclusions (Discussion). Conditioning on forcing (estimating effects on the deficit-to-impact *slope* rather than on levels) is how we break the collinearity between the 2010-onset megadrought and the treatment clock.

Hydrological drought. Daily streamflow from the DGA network (2000–2020), gathered by the CR2 (Center for Climate and Resilience Research, https://cr2.cl) is aggregated to the Standardized Streamflow Index at 12 months [SSI-12;[37];[38]], the regulated-flow analogue of SPEI-12 and the index in which meteorological deficits are re-expressed as hydrological drought, a propagation that reservoir operation is known to modify by lengthening and attenuating the transmitted signal[39,40], used as the outcome in the within-basin placebo (below). Gauges are classified as upstream or downstream of each dam from an SRTM digital elevation model. Because SSI standardizes each gauge on its own, for regulated gauges post-dam, flow distribution, the transmission slope tests distribution-relative propagation and would divide out any dam-induced variance compression; we therefore re-estimate every buffering model and the placebo on log 12-month mean flow, a volumetric semi-elasticity (percent of flow per SPEI unit) that no per-gauge normalization can compress, and test the compression premise itself, the deseasonalized relative flow variability by gauge class (Supplementary Table S19).

Ecological and land-use outcomes. (i) annual cropland area from MapBiomas Chile (https://chile.mapbiomas.org/) Collection 2 (30 m, 1999–2024); (ii) actual evapotranspiration from MODIS MOD16 (8-day, ~500 m, 2001–2024), for whole basins and for an orchard stratum delineated from the *Centro de Información de Recursos Naturales* (CIREN) and *Oficina de Estudios y Políticas Agrarias* (ODEPA) cadastre (https://icet.odepa.gob.cl/). Satellite vegetation and evapotranspiration signals track drought-driven losses of agricultural productivity in Chile closely enough to serve as impact outcomes[41].

Water rights. The DGA consolidated registry of water-use rights (*Derechos de Aprovechamiento de Aguas*), giving for each right its grant date, consumptive/non-consumptive type, surface/groundwater nature, use, allocated flow (harmonized to l/s), and UTM capture point. We assign consumptive rights to sub-watershed by point-in-polygon and build a per-basin annual panel of the cumulative allocated-rights stock, our direct demand-side outcome. Because the registry contains severe volume errors (a single physically impossible entry can dominate a basin's total, e.g. a caudal exceeding 200,000 m^3/s), the outlier-robust primary measure is rights *density* (count per 100 km^2); a volume series in which each individual consumptive flow is winsorized at the 99.5th percentile of the consumptive-flow distribution gives the same result, and the volume null is insensitive to the winsorization threshold across the 95th, 99th, 99.5th and 99.9th percentiles (permutation p from 0.39 to 0.21; Supplementary Table S8). Both are reported (Supplementary Fig. S6). Because volumes are extremely concentrated (the largest 1% of raw entries, dominated by the data errors, hold 99.7% of registry volume), we additionally report rank-based tests on the raw, uncapped volumes: the per-right volume distribution in dammed versus control basins, and a design rank contrast (ebal-weighted mean-rank difference) under both unit-level and watershed-block permutation, so no cap or outlier can drive the volume conclusion (Supplementary Table S18).

Snow water equivalent. ERA5-Land monthly snow-water-equivalent (SWE, 0.1°, 2000–2026), reduced to per-gauge and per-sub-watershed peak-SWE climatologies plus annual peak-SWE anomalies, to test the snowmelt confound on the placebo estimand itself (Supplementary Table S12): we refit the up- and downstream buffering models with a forcing-by-snowpack adjustment, within the low-snow half of units (below-median peak SWE), across early and late halves of the panel, and with a snow-year interaction that asks whether the apparent buffering strengthens in high-snow years. ERA5-Land underestimates absolute Andean snowpack, the same class of product caveat we apply to MOD16 evapotranspiration, so all snow terms enter standardized, which is insensitive to multiplicative bias; gauge-level sensitivity checks among undammed controls are likewise reported per standard deviation of peak SWE.

Groundwater levels. DGA well hydrographs of depth to water (2000–2021; 216 wells carry usable hydrographs nationally, of which 213 fall inside the matched basins and enter the bound; a Hampel filter removes measurement spikes), assigned to sub-watershed by point-in-polygon from well coordinates. Each well's megadrought (2010–2021) depletion rate is the ordinary lest square slope of depth to water on time (m yr^{-1}, positive = falling water table); per-basin values aggregate well rates by their outlier-robust median. These bound the groundwater-substitution confound as a matched dammed-versus-control Average Treatment Effect on the Treated (ATT) on the depletion rate (Supplementary Table S10).

Matching covariates. Baseline aridity (long-term mean annual precipitation divided by potential evapotranspiration, P/PET, over the 1991–2020 World Meteorological Organization normal; PET is the atmospheric evaporative demand and P/PET is used throughout), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per sub-watershed by zonal statistics. Because the 1991–2020 normal overlaps eleven megadrought years, baseline aridity could in principle absorb post-treatment variation; we verify the window is innocuous. Unit aridity recomputed on the strictly pre-drought 1991–2009 window is essentially identical to the full-window covariate (Pearson $r = 0.998$, Spearman $\rho = 0.999$ across the 265 matched units, and 99% of units keep the same aridity tercile used for the permutation strata), and the megadrought change in P/PET does not differ significantly between treated and weighted control basins (-0.04, $p = 0.10$, an order of magnitude smaller than the ~0.5 treated-control aridity gap the matching addresses), so the matching does not lean on drought-period or reservoir-influenced variation (Supplementary Table S13).

Matched control construction

We estimate the average treatment effect on the treated (ATT) using *entropy balancing* exact-matched within Köppen main group, balancing *log(area)* and *mean elevation* (see the archived analysis repository). Balancing on size and elevation rather than latitude is deliberate: latitude and the climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size (ESS) from 9 to 91 while balancing latitude for free.

Entropy balancing is a calibration estimator: given the weights, covariate balance is exact by construction and requires no independence assumption across units, so spatially autocorrelated covariates affect inference rather than balance, and inference is handled at the outcome stage with watershed-block permutation and clustered intervals (Supplementary Table S9); the exact matching within Köppen main group is itself a coarse spatial blocking, and latitude, though untargeted, balances as a by-product (Supplementary Fig. S5). The procedure retains 21 of 24 treated sub-watersheds (three high-Andes units fall outside local elevation common support) and achieves essentially perfect balance (standardized mean differences (SMD) for log(area) and elevation fall from 1.18 / 0.37 to 0.000, variance ratios 1), with an ESS 91 out of 244 in-window controls. The 244 in-window controls are the subset of the 398 clean controls that survives the common-support trim shared by every matching estimator: a control must lie in a Köppen main group containing a treated unit and within 250 m of that group's treated elevation range, and groups with fewer than ten such controls are dropped, so 154 clean controls fall outside this support and never enter the matching pool. Baseline aridity is a genuine confound (dammed basins are markedly more arid: median P/PET 0.26 vs 0.80 unweighted) but is *not* used as a balance target, because hard-balancing it collapses the ESS from 91 to 19; instead the weighting reduces the aridity imbalance to a residual SMD 0.17 (still above the 0.1 target), which the doubly-robust outcome model absorbs. We nonetheless report the strict hard-balanced fit (entropy balancing targeting log-aridity as well, SMD 0.000, ESS 19) as a primary sensitivity: with no parametric extrapolation across the regime gap at all, the streamflow, cropland, and water-rights results remain null, and even the *apparent* buffering vanishes, the direct signature of an aridity confound rather than a reservoir effect (Supplementary Table S17). At ESS 19 the hard-balanced fit is underpowered as a stand-alone estimator; its evidential value is agreement with the powered primary, not independent power. The parametric log-aridity adjustment is itself validated among undammed controls: their baseline SPEI-to-SSI slope varies smoothly and monotonically across aridity terciles (0.65 arid, 0.70 mid, 0.74 humid), with no regime break for a linear adjustment to misfit (Supplementary Table S20). Because this residual is real, the outcome-model adjustment for aridity is *load-bearing* rather than cosmetic; the gap it bridges is one of distributional density, not of support: all 21 dammed basins lie inside the control aridity range and inside the control 5th to 95th percentile band, and the control density is strictly positive across the treated support, so the adjustment interpolates within overlap rather than extrapolating beyond it (Supplementary Fig. S8). Accordingly, the reported cross-sectional ATTs use a *linear* log-aridity adjustment, adding a quadratic aridity term leaves the qualitative conclusions unchanged (archived robustness), and restricting to the aridity-overlap subset (all 21 treated basins and 158 of 244 controls fall inside it) leaves the streamflow transmission-slope gap essentially unchanged (-0.17 vs -0.18) while shrinking the water-rights ATT to a value that now straddles zero, so the parametric adjustment is not masking a true effect (Supplementary Table S6). To rule out that the nulls are artefacts of this parametric functional form, we additionally re-fit entropy balancing *within* the aridity-overlap subset and estimate every cross-sectional outcome with the weighting-only estimator, which carries no outcome-model aridity term and so performs no extrapolation across the regime gap. On this fully non-parametric matching specification, cropland-area expansion, orchard-ET buffering and the water-rights count all remain null (Supplementary Table S7); the winsorized water-rights volume, which behaves differently here, is treated under spatial inference below. Crucially, the study's decisive evidence, the within-basin upstream/downstream placebo, the within-region transmission-slope comparison, and randomization inference, does not use the outcome-model aridity adjustment at all, so the central nulls do not hinge on extrapolating across the aridity gap. Robustness to the matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-neighbour (Mahalanobis) matching on the identical common-support sample; entropy balancing and NN agree on all 21 treated, while CEM additionally prunes the 4 most extreme large/high basins, so all treatment effects are reported with and without those units.

Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the same sample, comprising weighting-only (ebal-weighted difference in means), regression-only (covariate-adjusted g-computation), and the headline *doubly-robust (DR)* estimator (ebal weights and an outcome model in log(area), elevation, and log(aridity)), so functional-form sensitivity is visible. Continuous covariates are centred at the treated mean so the treatment coefficient reads as the ATT; standard errors are HC3-robust (ebal weights treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (concentrated in exactly the arid basins where reservoirs sit), the outcome for the DR machinery is the per-basin *transmission slope*, the regression of an annual outcome (e.g. log evapotranspiration) on annual SPEI-12 over 2005–2024, a standardized, cross-basin drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope (more impact per unit deficit) in dammed basins.

Table 1 carries two cross-sectional matched ATTs, both level-change outcomes rather than transmission slopes: (i) *cropland-area expansion*, the change in MapBiomas cropland area per basin over the window (ha km^{-2} added, as Table 1 states), whose protection against differential megadrought exposure comes from the forcing-interacted difference-in-differences below; and (ii) *water-rights accrual*, described under Water rights above. A third cross-sectional outcome, *reservoir ET buffering*, applies the transmission-slope machinery to log evapotranspiration on SPEI over *irrigated orchard cells* (MOD16 ET on the CIREN/ODEPA cadastre footprint); because MOD16 underestimates ET over irrigated land, biasing that slope gap toward zero, the ET outcomes are reported with caveats in the Supplementary Information (Table S21) and carry no evidential weight.

Forcing-interacted difference-in-differences. On the observed area and ET panels we fit

$$y_{it} = \beta\,(\text{dammed}_i \times \text{SPEI}_{it}) + \gamma\,\text{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with sub-watershed (α_i) and year (τ_t) fixed effects and entropy-balancing weights. Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar year: the year fixed effects absorb the common megadrought shock, and β is the differential deficit-to-impact slope between dammed and matched-control basins, robust to the siting-in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability hypothesis predicts $\beta > 0$.

Equivalence testing and minimum detectable effect (MDE). Because the headline results are nulls, we move beyond failure-to-reject and bound the operational effect. For each null we report a two one-sided test (TOST): the effect is declared equivalent to zero if its 90% confidence interval lies within a negligible region of ±25% of the baseline (untreated) transmission slope. We are not aware of an established hydrological or operational standard defining a "negligible" reservoir effect on drought transmission, so we anchor the margin in the granularity of operational drought classification rather than choosing it arbitrarily. Standardized drought indices are managed in discrete severity categories each 0.5 standardized units wide[33,42]. Because the estimand is a slope, the absolute change it induces in the propagated index scales with the SPEI deficit, so the relevant test is at the most extreme drought actually observed. The largest SPEI-12 deficit in the panel is ≈ -2.0 (minimum -1.98), and 25% of the baseline transmission slope is $0.25 \times 0.585 \approx 0.146$ SSI per unit SPEI (0.585 is the untreated baseline of the intent-to-treat panel; the downstream-gauge panel baseline is 0.531, and both appear in Supplementary Table S2); the maximum absolute perturbation of the streamflow-drought index a 25% slope alteration can produce is therefore $0.146 \times 1.98 \approx 0.29$ SSI units, below the 0.5-unit width of a single severity category. A 25% slope change thus cannot move a basin across an operational drought-severity threshold even under the worst observed deficit, and is marginal for drought management. We stress that 25% of the transmission slope nonetheless corresponds to a non-trivial absolute water volume: translating the maximal margin perturbation (0.29 SSI units) through each treated downstream gauge's flow-to-SSI sensitivity gives a median of roughly 58 hm^3 over a 12-month season (interquartile range, IQR: 16 to 137 hm^3 across the 40 gauges, about 11% of mean annual flow; the linear sensitivity overstates the dry-tail volume, so these are upper-end figures; Supplementary Table S16). A volume of this order is genuinely material for local water management, so the margin is anchored to the operational drought-severity classification, not to volumetric negligibility, and the causal weight of the null rests on the within-basin placebo rather than on the equivalence verdict. We adopt ±25% as this deliberately strict, pre-specified margin and, because the equivalence verdict is nonetheless sensitive to the bound, also report the threshold-free MDE so that readers may apply any margin they consider material. The qualitative conclusion (only large effects are excluded; no outcome is equivalent at ±25%) does not depend on the exact margin. The minimum detectable effect (at 80% power, two-sided $\alpha = 0.05$) and the equivalence interval are built from the *standard deviation of the permutation-null distribution*, not the analytic standard error, since cluster-robust SEs over-reject at ~21 clusters (archived analysis code). To confirm the null is informative rather than underpowered, we additionally run a positive control. For each $\beta \in \{0, -0.15, -0.30, -0.45\}$ we set $\mathrm{SSI}^*_{it} = \mathrm{SSI}_{it} + \beta \times \mathrm{treat}_i \times \mathrm{SPEI}^c_{it}$, an additive perturbation on the standardized scale that injects a known buffering slope of magnitude β into treated units while leaving control observations and the panel's error structure untouched, then re-fit the full estimator and randomization inference on each perturbed panel (Supplementary Fig. S2 and Table S3). $\beta = 0$ reproduces the observed null exactly, recovery is one-to-one in β, and detection is required once $|\beta|$ exceeds the MDE, verifying that the design recovers an effect of policy-relevant size and correctly flags none when none is injected.

Siting-confound decomposition. To locate where any apparent buffering comes from, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects, what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights + fixed effects); (4) the design estimator with the baseline SPEI-to-outcome slope allowed to vary by Köppen region. It is important that these are distinct controls. The entropy balancing exact-matches *basin size and elevation* within Köppen main group; it does not balance the residual aridity gradient (which varies within B and C) nor impose a region-specific dose-response. The apparent buffering can therefore survive rungs (1) to (3), whose weighting leaves the within-region aridity gradient intact, and die only at rung (4), which lets each climate region have its own baseline transmission and so removes the between-region contrast that the matching does not. The collapse from (3) to (4) is the between-region aridity confound; the residual at (4) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated downstream reaches against physically unregulated upstream reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo's comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed. The linearity of that sensitivity is checked across the ~2000 m snowline with a piecewise fit among controls (no detectable break, interaction $p = 0.25$; Supplementary Table S20), and the snow regime itself is tested directly on the placebo estimand (Supplementary Table S12). To exclude that the pooled 2005–2024 slope averages an early-phase buffering (storage intact) into a late-phase failure (storage depleted), the buffering and placebo models are re-fit separately on the early (2010–2014) and late (2015–2024) megadrought phases, and the early-to-late change is tested as a treat $\times$ SPEI $\times$ phase interaction on the pooled megadrought panel (year fixed effects absorb the phase level; Supplementary Table S24).

Inference

With only ~21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is *randomization inference*: the treatment label is permuted among units within Köppen $\times$ aridity- tercile strata (999 permutations) and the observed statistic is compared against the resulting null distribution; cluster-robust confidence intervals are reported alongside for completeness. Because hydrological and land-use outcomes can be spatially autocorrelated, we assess residual spatial dependence for each outcome with Moran's I (great-circle distances, five-nearest-neighbour row-standardized weights, tested by 999 residual permutations). Where it is present, notably the water-rights outcomes ($I \approx 0.4$ to 0.6, $p = 0.001$; Supplementary Table S9), the unit-level stratified permutation can be anti-conservative because it does not preserve spatial structure. We therefore re-run the affected outcomes with a spatially-restricted permutation that swaps treatment at the watershed-block level, so co-located sub-watershed move together, and with watershed-block-clustered confidence intervals; this spatially-valid inference governs where the naive and spatial tests disagree (Supplementary Table S9). Parallel pre-trends are checked with a dynamic event study (dammed $\times$ year, reference 2009), and residual confounding is probed with an aridity placebo that relabels the most-arid control tercile as pseudo-treated.

Software and reproducibility

All analyses are implemented in R within a `targets` pipeline (matched-set construction, estimators, and figures are individual targets), using `WeightIt/cobalt` for balancing and diagnostics, `fixest` for fixed-effects regression, `sf/terra/exactextractr` for the spatial extractions, and `renv` for package pinning. Every result in the manuscript corresponds to a named pipeline target; no figure or table is produced by hand.

Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.21052149). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (SPEI-12 forcing and baseline aridity), MODIS MOD16 (evapotranspiration), MapBiomas Chile (land cover), the CR2 network (streamflow), ERA5-Land (snow water equivalent), SRTM (elevation), and the DGA reservoir, watershed, water-rights, and well-hydrograph registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

References

[1] K. E. Trenberth, A. Dai, G. van der Schrier, P. D. Jones, J. Barichivich, K. R. Briffa, J. Sheffield, Global warming and changes in drought, Nature Climate Change 4 (2014) 17–22.

[2] N. S. Diffenbaugh, D. L. Swain, D. Touma, Anthropogenic warming has increased drought risk in California, Proceedings of the National Academy of Sciences 112 (13), iSBN: 0027-8424, 1091-6490 (2015). doi:10.1073/pnas.1422385112.

[3] G. Di Baldassarre, N. Wanders, A. AghaKouchak, L. Kuil, S. Rangecroft, T. I. E. Veldkamp, M. Garcia, P. R. Van Oel, K. Breinl, A. F. Van Loon, Water shortages worsened by reservoir effects, Nature Sustainability 1 (11) (2018) 617–622. doi:10.1038/s41893-018-0159-0.
URL https://www.nature.com/articles/s41893-018-0159-0

[4] Y. Li, G. Zhao, G. H. Allen, H. Gao, Diminishing storage returns of reservoir construction, Nature Communications 14 (1) (2023) 3203. doi:10.1038/s41467-023-38843-5.
URL https://www.nature.com/articles/s41467-023-38843-5

[5] Z. Wang, L. Jiang, K. Nielsen, L. Wang, Reservoir Filling Up Problems in a Changing Climate: Insights From CryoSat-2 Altimetry, Geophysical Research Letters 51 (10) (2024) e2024GL108934. doi:10.1029/2024GL108934.
URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL108934

[6] E. Kellner, The controversial debate on the role of water reservoirs in reducing water scarcity, WIREs Water 8 (3) (2021) e1514. doi:10.1002/wat2.1514.
URL https://wires.onlinelibrary.wiley.com/doi/10.1002/wat2.1514

[7] J. Wu, X. Chen, H. Yao, L. Gao, Y. Chen, M. Liu, Non-linear relationship of hydrological drought responding to meteorological drought and impact of a large reservoir, Journal of Hydrology 551 (2017) 495–507. doi:10.1016/j.jhydrol.2017.06.029.
URL https://linkinghub.elsevier.com/retrieve/pii/S002216941730433X

[8] J. Wu, Z. Liu, H. Yao, X. Chen, X. Chen, Y. Zheng, Y. He, Impacts of reservoir operations on multi-scale correlations between hydrological drought and meteorological drought, Journal of Hydrology 563 (2018) 726–736. doi:10.1016/j.jhydrol.2018.06.053.
URL https://linkinghub.elsevier.com/retrieve/pii/S0022169418304761

[9] J. I. López-Moreno, S. M. Vicente-Serrano, S. Beguería, J. M. García-Ruiz, M. M. Portela, A. B. Almeida, Dam effects on droughts magnitude and duration in a transboundary basin: The Lower River Tagus, Spain and Portugal, Water Resources Research 45 (2) (2009) 2008WR007198. doi:10.1029/2008WR007198.
URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2008WR007198

[10] R. D. Garreaud, C. Alvarez-Garreton, J. Barichivich, J. P. Boisier, D. Christie, M. Galleguillos, C. LeQuesne, J. McPhee, M. Zambrano-Bigiarini, The 2010–2015 megadrought in central Chile: impacts on regional hydroclimate and vegetation, Hydrology and Earth System Sciences 21 (12) (2017) 6307–6327. doi:10.5194/hess-21-6307-2017.
URL https://hess.copernicus.org/articles/21/6307/2017/

[11] R. Garreaud, J. P. Boisier, C. Álvarez Garreton, D. Christie, T. Carrasco-Escaff, I. Vergara, R. O. Chávez, P. Aldunce, P. Camus, M. Suazo-Álvarez, M. Masiokas, G. Castro, A. Muñoz, M. Zambrano-Bigiarini, R. Fuster, L. Godoy, Hyperdroughts in central Chile: Drivers, Impacts and Projections (Mar. 2025). doi:10.5194/egusphere-2025-517.
URL https://egusphere.copernicus.org/preprints/2025/egusphere-2025-517/

[12] F. Zambrano, A. Vrieling, F. Meza, I. Duran-Llacer, F. Fernández, A. Venegas-González, N. Raab, D. Craven, From Drought to Aridification: Land-Cover Fingerprints of a Drying Chile, Earth's Future 13 (12) (2025) e2025EF006744. doi:10.1029/2025EF006744.
URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025EF006744

[13] J. P. Boisier, C. Alvarez-Garreton, R. R. Cordero, A. Damiani, L. Gallardo, R. D. Garreaud, F. Lambert, C. Ramallo, M. Rojas, R. Rondanelli, Anthropogenic drying in central-southern Chile evidenced by long-term observations and climate model simulations, Elementa 6 (1) (2018) 74. doi:10.1525/elementa.

328.
URL https://www.elementascience.org/article/10.1525/elementa.328/

[14] S. M. Vicente-Serrano, T. R. McVicar, D. G. Miralles, Y. Yang, M. Tomas-Burguera, Unraveling the influence of atmospheric evaporative demand on drought and its response to climate change, WIREs Climate Change (Dec. 2019). doi:10.1002/wcc.632.
URL https://onlinelibrary.wiley.com/doi/abs/10.1002/wcc.632

[15] P. Barría, I. B. Sandoval, C. Guzman, C. Chadwick, C. Alvarez-Garreton, R. Díaz-Vasconcellos, A. Ocampo-Melgar, R. Fuster, Water allocation under climate change, Elementa: Science of the Anthropocene 9 (1) (2021) 00131. doi:10.1525/elementa.2020.00131.
URL https://online.ucpress.edu/elementa/article/9/1/00131/117183/Water-allocation-under-climate-changeA-diagnosis

[16] B. Reade Malagueño, P. D'Odorico, *¿Libre de la Maleza Estatista?* Assessing Neoliberal Promises and Water Markets in Chile, Earth's Future 12 (10) (2024) e2024EF004963. doi:10.1029/2024EF004963.
URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024EF004963

[17] D. Rivera, A. Godoy-Faúndez, M. Lillo, A. Alvez, V. Delgado, C. Gonzalo-Martín, E. Menasalvas, R. Costumero, A. García-Pedrero, Legal disputes as a proxy for regional conflicts over water rights in Chile, Journal of Hydrology 535 (2016) 36–45. doi:10.1016/j.jhydrol.2016.01.057.
URL https://linkinghub.elsevier.com/retrieve/pii/S0022169416000755

[18] M. J. Webb, J. M. Winter, S. A. Spera, J. W. Chipman, E. C. Osterberg, Water, agriculture, and climate dynamics in central Chile's Aconcagua River Basin, Physical Geography 42 (5) (2021) 395–415, number: 5 Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/02723646.2020.1790719. doi:10.1080/02723646.2020.1790719.
URL https://doi.org/10.1080/02723646.2020.1790719

[19] C. Jordán, A. Engler, C. Bopp, P. M. Poortvliet, R. Jara-Rojas, Adaptation behavior to prolonged drought conditions: Farmers' responses to water scarcity in Central Chile, Environment, Development and Sustainability (Jun. 2025). doi:10.1007/s10668-025-06421-y.
URL https://link.springer.com/10.1007/s10668-025-06421-y

[20] J. T. Overpeck, B. Udall, Climate change and the aridification of North America, Proceedings of the

National Academy of Sciences 117 (22) (2020) 11856–11858. doi:10.1073/pnas.2006323117.
URL https://pnas.org/doi/full/10.1073/pnas.2006323117

[21] M. Ho, U. Lall, M. Allaire, N. Devineni, H. H. Kwon, I. Pal, D. Raff, D. Wegner, The future role of dams in the United States of America, Water Resources Research 53 (2) (2017) 982–998. doi:10.1002/2016WR019905.
URL https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2016WR019905

[22] C. Alvarez-Garreton, J. P. Boisier, R. Garreaud, J. Seibert, M. Vis, Progressive water deficits during multiyear droughts in basins with long hydrological memory in Chile, Hydrology and Earth System Sciences 25 (1) (2021) 429–446. doi:10.5194/hess-25-429-2021.
URL https://hess.copernicus.org/articles/25/429/2021/

[23] M. Schilstra, W. Wang, P. R. Van Oel, J. Wang, H. Cheng, The effects of reservoir storage and water use on the upstream–downstream drought propagation, Journal of Hydrology 631 (2024) 130668. doi:10.1016/j.jhydrol.2024.130668.
URL https://linkinghub.elsevier.com/retrieve/pii/S0022169424000623

[24] C. Lee, S. Kang, S. Kim, Drought resistance of dams based on propagation analysis: A case study of various multipurpose dams in South Korea, Journal of Hydrology: Regional Studies 63 (2026) 103106. doi:10.1016/j.ejrh.2026.103106.
URL https://linkinghub.elsevier.com/retrieve/pii/S2214581826000042

[25] K. Y. Kim, H. Rajaram, V. Lakshmi, Observing decreasing snow cover and increasing surface temperature across the Andes with remotely sensed and reanalysis data, Environmental Research Communications 7 (2025) 021009. doi:10.1088/2515-7620/adb382.

[26] E. Macpherson, C. Clark, P. W. Salazar, N. Baird, A. Akhtar-Khavari, E. Challies, Evolving rights to (and of) water in Chile: a case for relationship-based water law and governance, The International Journal of Human Rights (2023) 1–26doi:10.1080/13642987.2023.2266719.
URL https://www.tandfonline.com/doi/full/10.1080/13642987.2023.2266719

[27] R. Q. Grafton, J. Williams, C. J. Perry, F. Molle, C. Ringler, P. Steduto, B. Udall, S. A. Wheeler, Y. Wang, D. Garrick, R. G. Allen, The paradox of irrigation efficiency, Science 361 (6404) (2018) 748–750, publisher: American Association for the Advancement of Science. doi:10.1126/science.aat9314.

[28] B. R. Scanlon, S. Fakhreddine, A. Rateb, I. De Graaf, J. Famiglietti, T. Gleeson, R. Q. Grafton, E. Jobbagy, S. Kebede, S. R. Kolusu, L. F. Konikow, D. Long, M. Mekonnen, H. M. Schmied, A. Mukherjee, A. MacDonald, R. C. Reedy, M. Shamsudduha, C. T. Simmons, A. Sun, R. G. Taylor, K. G. Villholth, C. J. Vörösmarty, C. Zheng, Global water resources and the role of groundwater in a resilient water future, Nature Reviews Earth & Environment 4 (2) (2023) 87–101. doi:10.1038/s43017-022-00378-6.
URL https://www.nature.com/articles/s43017-022-00378-6

[29] R. Rivera-Vidal, J. L. Arumí, O. Melo, V. Delgado, V. Parra, A. Stehr, L. Daniele, Managed aquifer recharge implementation challenges: Lessons from Chile's water-scarce regions, Groundwater for Sustainable Development 31 (2025) 101502. doi:10.1016/j.gsd.2025.101502.
URL https://linkinghub.elsevier.com/retrieve/pii/S2352801X25000992

[30] M. Taucare, B. Viguier, R. Figueroa, L. Daniele, The alarming state of Central Chile's groundwater resources: A paradigmatic case of a lasting overexploitation, Science of The Total Environment 906 (2024) 167723. doi:10.1016/j.scitotenv.2023.167723.
URL https://linkinghub.elsevier.com/retrieve/pii/S0048969723063507

[31] Á. Ayala, E. Muñoz-Castro, D. Farinotti, D. Farías-Barahona, P. A. Mendoza, S. MacDonell, J. McPhee, X. Vargas, F. Pellicciotti, Less water from glaciers during future megadroughts in the Southern Andes, Communications Earth & Environment (2025). doi:10.1038/s43247-025-02845-6.

[32] J. Götte, M. I. Brunner, Hydrological Drought-To-Flood Transitions Across Different Hydroclimates in the United States, Water Resources Research 60 (7) (2024) e2023WR036504. doi:10.1029/2023WR036504.
URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023WR036504

[33] S. M. Vicente-Serrano, S. Beguería, J. I. López-Moreno, A multiscalar drought index sensitive to global warming: The standardized precipitation evapotranspiration index, Journal of Climate 23 (7) (2010) 1696–1718. doi:10.1175/2009JCLI2909.1.
URL http://dx.doi.org/10.1175/2009JCLI2909.1

[34] F. Zambrano, B. Wardlow, T. Tadesse, M. Lillo-Saavedra, O. Lagos, Evaluating satellite-derived long-term historical precipitation datasets for drought monitoring in Chile, Atmospheric Research 186 (2017) 26–42, publisher: Elsevier. doi:10.1016/j.atmosres.2016.11.006.
URL https://www.sciencedirect.com/science/article/pii/S0169809516305865

[35] F. J. Meza, Recent trends and {ENSO} influence on droughts in {N}orthern {C}hile: {A}n application of the {S}tandarized {P}recipitation {E}vapotranspiration {I}ndex, Weather and Climate Extremes 1 (2013) 51–58.

[36] M. Oertel, F. J. Meza, J. Gironás, Observed trends and relationships between ENSO and standardized hydrometeorological drought indices in central Chile, Hydrological Processes 34 (2) (2020) 159–174. doi:10.1002/hyp.13596.
URL https://onlinelibrary.wiley.com/doi/10.1002/hyp.13596

[37] S. M. Vicente-Serrano, J. I. López-Moreno, S. Beguería, J. Lorenzo-Lacruz, C. Azorin-Molina, E. Morán-Tejeda, Accurate computation of a streamflow drought index, J. Hydrol. Eng. 17(2) (2012) 318–332.
URL http://hdl.handle.net/10261/49224

[38] F. Lema, P. A. Mendoza, N. A. Vásquez, N. Mizukami, M. Zambrano-Bigiarini, X. Vargas, Technical note: What does the Standardized Streamflow Index actually reflect? Insights and implications for hydrological drought analysis, Hydrology and Earth System Sciences 29 (8) (2025) 1981–2002. doi:10.5194/hess-29-1981-2025.
URL https://hess.copernicus.org/articles/29/1981/2025/

[39] A. F. Van Loon, Hydrological drought explained, WIREs Water 2 (4) (2015) 359–392. doi:10.1002/wat2.1085.
URL https://wires.onlinelibrary.wiley.com/doi/10.1002/wat2.1085

[40] W. Wang, M. W. Ertsen, M. D. Svoboda, M. Hafeez, Propagation of drought: From meteorological drought to agricultural and hydrological drought, Advances in MeteorologyArXiv: 6547209 (2016). doi:10.1155/2016/6547209.

[41] F. Zambrano, A. Vrieling, A. Nelson, M. Meroni, T. Tadesse, Prediction of drought-induced reduction of agricultural productivity in Chile from MODIS, rainfall estimates, and climate oscillation indices, Remote Sensing of Environment 219 (2018) 15–30, publisher: Elsevier. doi:10.1016/j.rse.2018.10.006.
URL https://www.sciencedirect.com/science/article/pii/S0034425718304541

[42] T. B. McKee, N. J. Doesken, J. Kleist, The relationship of drought frequency and duration to time scales, Proceedings of the Ninth Conference on Applied Climatology, American Meteorological Society 17 (22) (1993) 179–184.